

P·E·D·W Advanced

One hidden layer, real weight matrices - same four moves - YOUR TURN

THE SETUP - DAY D13 OF THE GOLF LOGBOOK

D13 said: hot, dry, calm ...and they PLAYED! (so this time the target is 1)

layer_0 =

Temp	Humidity	Wind
1	0	0

target y =

Play?
1

The weight matrices (random start, seed(1), rounded to 2 decimals - your hand math will check exactly):

weights_0_1 =

	node1	node2	node3	node4
Temp	-0.17	0.44	-1.00	-0.40
Humidity	-0.71	-0.82	-0.63	-0.31
Wind	-0.21	0.08	-0.16	0.37

weights_1_2 =

node1	-0.59
node2	0.76
node3	-0.95
node4	0.34

$\alpha = 0.1$

(learning rate) · $3 \times 4 = 12$ weights into the hidden layer, 4 more to the output: 16 total

TWO TINY HANDCRAFTED FUNCTIONS - THIS IS ALL THE MAGIC

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def relu(x):
    return (x > 0) * x

def relu2deriv(output):
    return output > 0
```

relu: a negative number gets recoded to 0.
(positive numbers pass straight through)

relu2deriv: THE MASK. 1 if the node fired,
0 if it slept. Sleeping nodes learn nothing.

Forward: $\text{layer}_1 = \text{relu}(\text{layer}_0 \cdot W_{01})$ then $\text{prediction} = \text{layer}_1 \cdot W_{12}$ (no relu at the output!)

Backward: the delta flows back through the SAME weights - and relu2deriv silences the nodes that never fired.

THE FLIGHT PLAN (same P·E·D·W, just with matrices)

p.2 FORWARD: layer_1 before relu $\rightarrow$ relu $\rightarrow$ prediction then E and D

p.3 BACKWARD: update weights_1_2, build layer_1_delta (the mask!), update weights_0_1

FORWARD PASS + E + D · YOUR TURN

P - step 1: layer_1 (before relu) = $\text{layer}_0 \cdot W_{01}$

One dot product per hidden node - walk the columns of W_{01} with $x = [1, 0, 0]$:

node1: $(1 \times -0.17) + (0 \times -0.71) + (0 \times -0.21) =$

node2: $(1 \times 0.44) + (0 \times -0.82) + (0 \times 0.08) =$

node3: $(1 \times -1.00) + (0 \times -0.63) + (0 \times -0.16) =$

node4: $(1 \times -0.40) + (0 \times -0.31) + (0 \times 0.37) =$

layer_1 (pre-relu) =

node1	node2	node3	node4

P - step 2: $\text{layer}_1 = \text{relu}(\text{layer}_1)$

Negative? Recode to 0. Positive? Pass through.

$\text{layer}_1 =$

node1	node2	node3	node4

P - step 3: prediction = $\text{layer}_1 \cdot W_{12}$ then E and D

P $(0 \times -0.59) + (\text{ } \times 0.76) + (0 \times -0.95) + (0 \times 0.34) =$

E $(1 - \text{ })^2 =$

D $\text{ } - 1 =$

Is D positive or negative? Did we over- or under-estimate? Which direction must the messenger weights move?

BACKWARD PASS - THE W IN P·E·D·W · YOUR TURN

W (part a): $\text{weight_delta_1_2} = \text{layer_1} \times D$ then update W12

Each hidden node's activation times the delta. Keep 4 decimals.

	weight_delta_1_2		new W12	
node1:	$0 \times -0.6656 =$		$-0.59 - 0.1 \times 0 =$	
node2:	$0.44 \times -0.6656 =$		$0.76 - 0.1 \times (-0.2929) =$	
node3:	$0 \times -0.6656 =$		$-0.95 - 0.1 \times 0 =$	
node4:	$0 \times -0.6656 =$		$0.34 - 0.1 \times 0 =$	

How many of the four weights actually move? Why those?

W (part b): $\text{layer_1_delta} = D \times W12$...then APPLY THE MASK

Send the miss backward through the same weights: $D \times$ each W12 value. Keep 4 decimals.

	node1	node2	node3	node4
before the mask =				

$\times \text{relu2deriv}(\text{layer_1}) = [0, 1, 0, 0] \rightarrow$

layer_1_delta =				
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W (part c): $\text{weight_delta_0_1} = \text{layer_0}^T \times \text{layer_1_delta}$ then update W01

Outer product: each INPUT row times layer_1_delta. Inputs Humidity and Wind were 0 - whole rows of zeros!

	node1	node2	node3	node4
Temp				
Hum				
Wind				

	node1	node2	node3	node4
new W01 =				

Update rule per cell: $\text{new} = \text{old} - 0.1 \times \text{delta}$. How many of the 16 weights actually change?

CHECK YOURSELF: $P = 0.3344$ $D = -0.6656$ $\text{new W12 node2} = 0.7893$ $\text{new W01 (Temp} \rightarrow \text{node2)} = 0.4906$

Everything else stays frozen. If more than 2 of your weights moved, hunt down which zero you dropped.